

Relativistic Quantum Mechanics I: Four-dimensional Spinors

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1 Three-dimensional Spinors

1.1 Wave Function of a Spin $\frac{1}{2}$ Particle

Consider a spin $\frac{1}{2}$ particle like an electron, its state $|\psi(\mathbf{r}, \sigma)\rangle$ depends on four variables: the particle's coordinate $\mathbf{r}$ and the discrete spin variable $\sigma \equiv s_z/\hbar$. Let $|\alpha\rangle$ and $|\beta\rangle$ denote the states where $\sigma = \frac{1}{2}$ and $-\frac{1}{2}$, then we can write $|\psi(\mathbf{r}, \sigma)\rangle$ as

$$|\psi(\mathbf{r}, \sigma)\rangle = |\alpha\rangle\langle\alpha|\psi(\mathbf{r}, \sigma)\rangle + |\beta\rangle\langle\beta|\psi(\mathbf{r}, \sigma)\rangle \equiv |\alpha\rangle\psi_\alpha(\mathbf{r}) + |\beta\rangle\psi_\beta(\mathbf{r}) \quad (1.1)$$

where $\psi_\alpha(\mathbf{r}) = \langle\alpha|\psi(\mathbf{r}, \sigma)\rangle$ and $\psi_\beta(\mathbf{r}) = \langle\beta|\psi(\mathbf{r}, \sigma)\rangle$. **The wave function of this particle thus has two components, i.e., ψ_α and ψ_β .** Clearly, $\int |\psi_\alpha(\mathbf{r})|^2 d\mathbf{r}$ is the probability that the particle has spin α and $\int |\psi_\beta(\mathbf{r})|^2 d\mathbf{r}$ is the probability that the particle has spin β , whereas $\sum_\sigma |\psi_\sigma(\mathbf{r})|^2$ gives the probability to find the particle at position $\mathbf{r}$. $|\psi(\mathbf{r}, \sigma)\rangle$ can be expressed in a two-row column matrix:

$$|\psi(\mathbf{r}, \sigma)\rangle \equiv \begin{pmatrix} \psi_\alpha(\mathbf{r}) \\ \psi_\beta(\mathbf{r}) \end{pmatrix} \quad (1.2)$$

Next, let's consider the transformation of the particle's wave function under the rotation in 3D space. **Here we consider the rotation of the coordinate system instead of rotation of the particle itself.** When the coordinate system is rotated, we can view it as the particle rotated in the opposite direction. For simplicity, consider the rotation of the yz plane **counterclockwise** by ϕ (the particle is rotated clockwise by $-\phi$), the rotation operator is given by

$$\hat{R} = \exp\left(\frac{i\hat{s}_x\phi}{\hbar}\right) \quad (1.3)$$

The state after the rotation is

$$|\psi'(\mathbf{r}, \sigma)\rangle = \hat{R}|\psi(\mathbf{r}, \sigma)\rangle \quad (1.4)$$

$\psi_\alpha(\mathbf{r})$ and $\psi_\beta(\mathbf{r})$ are scalar functions and are thus invariant under rotation. It's easy to show that

$$\begin{aligned} |\psi'(\mathbf{r}, \sigma)\rangle &= \left[\cos \frac{\phi}{2} \psi_\alpha(\mathbf{r}) + i \sin \frac{\phi}{2} \psi_\beta(\mathbf{r}) \right] |\alpha\rangle \\ &\quad + \left[i \sin \frac{\phi}{2} \psi_\alpha(\mathbf{r}) + \cos \frac{\phi}{2} \psi_\beta(\mathbf{r}) \right] |\beta\rangle \end{aligned} \quad (1.5)$$

Or equivalently,

$$\begin{pmatrix} \psi'_\alpha(\mathbf{r}) \\ \psi'_\beta(\mathbf{r}) \end{pmatrix} = \begin{pmatrix} \cos \frac{\phi}{2} & i \sin \frac{\phi}{2} \\ i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix} \begin{pmatrix} \psi_\alpha(\mathbf{r}) \\ \psi_\beta(\mathbf{r}) \end{pmatrix} \quad (1.6)$$

In general, for a rotation around arbitrary unit vector $\mathbf{n}$ by ϕ ,

$$\begin{pmatrix} \psi'_\alpha(\mathbf{r}) \\ \psi'_\beta(\mathbf{r}) \end{pmatrix} = \begin{pmatrix} \cos \frac{\phi}{2} + i n_z \sin \frac{\phi}{2} & (i n_x + n_y) \sin \frac{\phi}{2} \\ (i n_x - n_y) \sin \frac{\phi}{2} & \cos \frac{\phi}{2} - i n_z \sin \frac{\phi}{2} \end{pmatrix} \begin{pmatrix} \psi_\alpha(\mathbf{r}) \\ \psi_\beta(\mathbf{r}) \end{pmatrix} \quad (1.7)$$

The two components of the wave function after rotation are linear combinations of the two components of the original function, and the transformation matrix is unitary and unimodular (i.e., its determinant equals to 1). A quantity with these properties is called a **spinor**.

Eq. (1.7) has the following implications. Firstly, when $\phi = 2\pi$, we have

$$\begin{pmatrix} \psi'_\alpha(\mathbf{r}) \\ \psi'_\beta(\mathbf{r}) \end{pmatrix}_{\phi=2\pi} = - \begin{pmatrix} \psi_\alpha(\mathbf{r}) \\ \psi_\beta(\mathbf{r}) \end{pmatrix} \quad (1.8)$$

Thus, **the wave function of a spin $\frac{1}{2}$ particle changes sign after rotation by 2π . A rotation by 4π is needed to revert to the original wave function.** This is one of the special properties of a spinor. Secondly, during rotation, the probability density of the particle $\sum_\sigma |\psi'_\sigma(\mathbf{r})|^2 = \sum_\sigma |\psi_\sigma(\mathbf{r})|^2$ is invariant, as it should be. In other word, $\sum_\sigma |\psi_\sigma(\mathbf{r})|^2$ **is a scalar**.

The transformation matrix in Eq. (1.7) has three independent parameters (four parameters n_x, n_y, n_z, ϕ and a constraint $n_x^2 + n_y^2 + n_z^2 = 1$). This is consistent with that the rotation in 3D space has three degrees of freedom (three Euler angles). Therefore, **the transformation matrices are representations of the 3D rotation group, SO(3). These unitary and unimodular matrices form another group named SU(2), which is homomorphic to the SO(3) group.** For one roataiotn angle ϕ , we can construct two unitary and unimodular matrices for ϕ and $\phi + 2\pi$, respectively. The homomorphism of SU(2) to SO(3) is thus two-to-one.

1.2 Three-dimensional Spinors

A quantity ξ having two components, ξ^1 and ξ^2 , which are transformed as

$$\begin{pmatrix} \xi'^1 \\ \xi'^2 \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \xi^1 \\ \xi^2 \end{pmatrix} \quad (1.9)$$

with a constraint $\alpha\delta - \beta\gamma = 1$ under 3D spatial rotation is called a **spinor of rank one** (more specifically, a three-dimensional spinor or 3-spinor of rank one). This transformation is called a **binary transformation**. ξ^1 and ξ^2 are the **contravariant components** of the spinor. **The wave function of a spin $\frac{1}{2}$ particle is a spinor* and its components ψ_α and ψ_β are contravariant components of the spinor.**

*We will omit the rank for spinors of rank one hereafter.

Using Eq. (1.9), for two spinors ξ and η , under spatial rotation, the expression

$$\xi'^1 \eta'^2 - \xi'^2 \eta'^1 = (\alpha\delta - \beta\gamma)(\xi^1 \eta^2 - \xi^2 \eta^1) = \xi^1 \eta^2 - \xi^2 \eta^1 \quad (1.10)$$

is invariant, because $\alpha\delta - \beta\gamma = 1$. Therefore we can construct a scalar, $\xi^1 \eta^2 - \xi^2 \eta^1$, from two spinors.

The **covariant components** of the spinor, ξ_1 and ξ_2 , can be obtained via

$$\xi_\alpha = g_{\alpha\beta} \xi^\beta \quad (1.11)$$

Here we use Greek letters $\alpha, \beta, \dots$ as indices of spinors, and the Einstein summation convention is applied. $g_{\alpha\beta}$ is the “**metric spinor**” in the spinor space,

$$g_{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (1.12)$$

Therefore,

$$\xi_1 = \xi^2, \quad \xi_2 = -\xi^1 \quad (1.13)$$

Substituting this relation in Eq. (1.9), we obtain the transformation of the covariant components of a spinor under spatial rotation:

$$\begin{pmatrix} \xi'_1 \\ \xi'_2 \end{pmatrix} = \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \quad (1.14)$$

The covariant components of the “metric spinor” are $g^{\alpha\beta}$,

$$\xi^\beta = \xi_\alpha g^{\alpha\beta}, \quad g_{\alpha\beta} = g^{\alpha\beta} \quad (1.15)$$

The **scalar product** of two spinors can thus be written using the covariant and contravariant components in the same manner as in tensor algebra,

$$\xi^\alpha \eta_\alpha = \xi^1 \eta_1 + \xi^2 \eta_2 = \xi^1 \eta^2 - \xi^2 \eta^1 = -\xi_2 \eta^2 + \xi_1 \eta^1 = -\xi_\alpha \eta^\alpha \quad (1.16)$$

We notice immediately that $\xi^\alpha \xi_\alpha = -\xi_\alpha \xi^\alpha = 0$.

Next, we consider the transformation of the complex conjugate of a spinor, ξ^* , under spatial rotation. Taking complex conjugate of Eq. (1.9), we have

$$\begin{pmatrix} \xi'^{1*} \\ \xi'^{2*} \end{pmatrix} = \begin{pmatrix} \alpha^* & \beta^* \\ \gamma^* & \delta^* \end{pmatrix} \begin{pmatrix} \xi^{1*} \\ \xi^{2*} \end{pmatrix} \quad (1.17)$$

Since the expression $\xi^{1*} \xi^1 + \xi^{2*} \xi^2$ corresponds to the probability density of a particle and must be a scalar, $\xi^{\alpha*}$ **must be transformed as ξ_α under spatial rotation**. This is denoted using “ $\sim$ ” symbol:

$$\xi^{\alpha*} \sim \xi_\alpha \quad (1.18)$$

Comparing Eq. (1.17) and Eq. (1.14), obviously,

$$\boxed{\alpha = \delta^*, \quad \beta = -\gamma^*} \quad (1.19)$$

Eqs. (1.9) and (1.14) can then be rewritten as

$$\boxed{\begin{pmatrix} \xi'^1 \\ \xi'^2 \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix} \begin{pmatrix} \xi^1 \\ \xi^2 \end{pmatrix}, \quad \begin{pmatrix} \xi'_1 \\ \xi'_2 \end{pmatrix} = \begin{pmatrix} \alpha^* & \beta^* \\ -\beta & \alpha \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}} \quad (1.20)$$

The transformation matrices are now unitary. They also contain three free parameters (the real and imaginary parts of α and β with a constraint $|\alpha|^2 + |\beta|^2 = 1$). Comparing this equation with Eq. (1.7), we have

$$\text{Re } \alpha = \cos \frac{\phi}{2}, \quad \text{Im } \alpha = n_z \sin \frac{\phi}{2} \quad (1.21)$$

$$\text{Re } \beta = n_y \sin \frac{\phi}{2}, \quad \text{Im } \beta = n_x \sin \frac{\phi}{2} \quad (1.22)$$

1.3 Spinors of Higher Rank and Particles with Arbitrary Spin

A **spinor of rank two** is a quantity $\zeta^{\alpha\beta}$ whose four components ($\alpha, \beta = 1, 2$) are transformed as the product of two spinors of rank one $\xi^\alpha \eta^\beta$ under spatial rotation. $\zeta^{\alpha\beta}$ are the contravariant components of the spinor of rank two. The covariant and mix components can be obtained via contraction with the “metric spinor”,

$$\zeta_\alpha{}^\beta = g_{\alpha\gamma} \zeta^{\gamma\beta}, \quad \zeta^\alpha{}_\beta = g_{\gamma\beta} \zeta^{\alpha\gamma}, \quad \zeta_{\alpha\beta} = g_{\alpha\gamma} g_{\beta\delta} \zeta^{\gamma\delta} \quad (1.23)$$

Note that $g_{\alpha\beta}$ itself is an antisymmetrical spinor of rank two. Spinors with higher rank can be similarly defined. Multiplication of two spinors of rank n and m gives a spinor of rank $n + m$, where as contraction of a spinor of rank n with respect to m pairs of indices gives a spinor of rank $n - 2m$.

Similar to the spinors of rank one, the complex conjugation of the contravariant components of a spinor of rank n , $\zeta^{\dots\alpha\beta\dots*}$ are transformed as the covariant components $\zeta_{\dots\alpha\beta\dots}$ and *vice versa*. Therefore, the sum of the norm square of the components of a spinor of rank n , $\sum_{\dots,\alpha,\beta,\dots} |\zeta^{\dots\alpha\beta\dots}|^2$ is invariant under spatial rotation.

A **symmetrical spinor of rank n** is a spinor which is symmetrical with respect to all its indices. Since a spinor of rank n is transformed as a product of n spinors of rank one, according to Eq. (1.16), interchanging the upper and lower indices in a contraction changes the sign, we can see that contraction of a symmetrical spinor with respect to any pair of indices gives 0. In general, a spinor of rank n has 2^n components. However, **a symmetric spinor of rank n only has $n + 1$ independent components**, because all the components whose indices contain l ones and $n - l$ twos are the same. For a given l , there are $\frac{n!}{l!(n-l)!}$ components and $\frac{(n)!}{l!(n-l)!} - 1$ constraints. There are therefore in total $\sum_{l=0}^n \left(\frac{n!}{l!(n-l)!} - 1 \right) = 2^n - n - 1$ constraints.

The wave function of a spin s particle has $2s + 1$ components ψ_σ where $\sigma = s, s - 1, \dots, -s$. On the other hand, this particle can be viewed as a composite of $2s$ spin $\frac{1}{2}$ particles. The wave

function of $2s$ spin $\frac{1}{2}$ particles is the product of $2s$ spinors of rank one, $\xi^\alpha \eta^\beta \dots$. Obviously, this wave function should be symmetrical with respect to all indices. Therefore, **the wave function of a spin s particle is a symmetrical spinor of rank $2s$** . This symmetrical spinor also has $2s + 1$ independent components, as discussed above.

We can find the relation between these independent contravariant components and ψ_σ . For a spin s particle with a given σ , it corresponds to the $2s$ spin $\frac{1}{2}$ particle system where $s + \sigma$ particles have $\sigma_i = \frac{1}{2}$ and $s - \sigma$ particles have $\sigma_i = -\frac{1}{2}$. Thus

$$\psi_\sigma \xleftrightarrow{\text{corresponds to}} \zeta \underbrace{11\dots 1}_{s+\sigma} \underbrace{22\dots 2}_{s-\sigma} \quad (1.24)$$

There are $\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}$ components of the spinor of rank $2s$ whose indices contain $s + \sigma$ ones and $s - \sigma$ twos. Meanwhile, the probability of finding the particle at a given point in space is $\sum_\sigma |\psi_\sigma|^2 = \sum_{\alpha, \beta, \dots} |\zeta^{\alpha\beta\dots}|^2$. Therefore

$$\psi_\sigma = \sqrt{\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}} \zeta \underbrace{11\dots 1}_{s+\sigma} \underbrace{22\dots 2}_{s-\sigma} \quad (1.25)$$

For the covariant components, note that from Eq. (1.13),

$$\zeta_{\underbrace{11\dots 1}_{s+\sigma} \underbrace{22\dots 2}_{s-\sigma}} = (-1)^{s-\sigma} \zeta \underbrace{11\dots 1}_{s-\sigma} \underbrace{22\dots 2}_{s+\sigma} \quad (1.26)$$

Thus

$$\psi_{-\sigma} = (-1)^{s-\sigma} \sqrt{\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}} \zeta_{\underbrace{11\dots 1}_{s+\sigma} \underbrace{22\dots 2}_{s-\sigma}} \quad (1.27)$$

Combining the relations of ψ_σ and $\psi_{-\sigma}$ with the contravariant and covariant components, we have

$$\zeta^{\alpha\beta\dots} \zeta_{\alpha\beta\dots} = \sum_\sigma (-1)^{s-\sigma} \psi_\sigma \psi_{-\sigma} \quad (1.28)$$

1.4 Relation between Spinors and Tensors

The spatial rotation transforms the $2s + 1$ components of a symmetrical spinor of rank $2s$ to linear combinations of these components via binary transformation. In 3D space, an **irreducible symmetrical tensor of rank s** (s is an integer) is a tensor which is symmetrical with respect to all its indices, and contraction with respect to any pairs of indices gives 0. One can show that an irreducible symmetrical tensor of rank s also only has $2s + 1$ independent components. Spatial rotation transforms these components into their linear combinations via matrix multiplication with 3×3 rotation matrices. **Since both the binary transformation matrices and the 3×3 rotation matrices are representations of the 3D rotation group, there must be a correspondence between a symmetrical spinor of rank $2s$ and an irreducible symmetrical tensor of rank s when s is an integer.**

We first consider the case where $s = 1$. The contravariant components of a symmetrical spinor ζ of rank two must satisfy the condition,

$$\zeta^{12} = \zeta^{21} \quad (1.29)$$

which is equivalent to (using Eq. (1.13)):

$$\zeta^1_1 = -\zeta^2_2 \quad (1.30)$$

thus $\zeta^\alpha_\alpha = 0$. In other words, **the mixed components of a symmetrical spinor of rank two form a 2×2 matrix with zero trace**. Such matrix can be constructed as linear combination of the **Pauli matrices**:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (1.31)$$

$$\zeta^\alpha_\beta = a_x \sigma_x + a_y \sigma_y + a_z \sigma_z = \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix} \quad (1.32)$$

where a_x, a_y, a_z are expansion coefficients. Spatial rotation transforms ζ^α_β to ζ'^α_β , which can also be expressed as linear combination of the Pauli matrices with new coefficients a'_x, a'_y, a'_z .

We need to find out the relations between a'_x, a'_y, a'_z and a_x, a_y, a_z . Since ζ^α_β is transformed as ξ^α_η , these coefficients must satisfy the equation

$$\begin{pmatrix} a'_z & a'_x - ia'_y \\ a'_x + ia'_y & -a'_z \end{pmatrix} = \mathbf{U} \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix} \mathbf{U}^H \quad (1.33)$$

where

$$\mathbf{U} = \begin{pmatrix} \cos \frac{\phi}{2} + in_z \sin \frac{\phi}{2} & (in_x + n_y) \sin \frac{\phi}{2} \\ (in_x - n_y) \sin \frac{\phi}{2} & \cos \frac{\phi}{2} - in_z \sin \frac{\phi}{2} \end{pmatrix} \quad (1.34)$$

and $\mathbf{U}^H$ is the Hermitian adjoint of $\mathbf{U}$. Solving this equation, we have

$$\begin{pmatrix} a'_x \\ a'_y \\ a'_z \end{pmatrix} = \mathbf{R} \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} \quad (1.35)$$

where

$$\mathbf{R} = \begin{pmatrix} \cos \phi + n_x^2(1 - \cos \phi) & n_x n_y(1 - \cos \phi) + n_z \sin \phi & n_x n_z(1 - \cos \phi) - n_y \sin \phi \\ n_x n_y(1 - \cos \phi) - n_z \sin \phi & \cos \phi + n_y^2(1 - \cos \phi) & n_y n_z(1 - \cos \phi) + n_x \sin \phi \\ n_x n_z(1 - \cos \phi) + n_y \sin \phi & n_y n_z(1 - \cos \phi) - n_x \sin \phi & \cos \phi + n_z^2(1 - \cos \phi) \end{pmatrix} \quad (1.36)$$

is a 3×3 rotation matrix for the rotation around unit vector $\mathbf{n}$ by ϕ . Therefore, $\mathbf{a} = (a_x, a_y, a_z)$ is **a three-dimensional vector**. We can thus conclude that **a symmetrical spinor of rank two**

corresponds to a three-dimensional vector. Let $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, the relation between their components can be expressed as

$$\zeta^\alpha{}_\beta = \boldsymbol{\sigma} \cdot \mathbf{a} = \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix} \quad (1.37)$$

For the contravariant components,

$$\zeta^{\alpha\beta} = \begin{pmatrix} -a_x + ia_y & a_z \\ a_z & a_x + ia_y \end{pmatrix} \quad (1.38)$$

Conversely, We can construct a vector from a spinor of rank two via

$$\begin{aligned} a_x &= \frac{1}{2}(\zeta^1{}_2 + \zeta^2{}_1) = \frac{1}{2}(\zeta^{22} - \zeta^{11}) \\ a_y &= \frac{i}{2}(\zeta^1{}_2 - \zeta^2{}_1) = -\frac{i}{2}(\zeta^{11} + \zeta^{22}) \\ a_z &= \frac{1}{2}(\zeta^1{}_1 - \zeta^2{}_2) = \frac{1}{2}(\zeta^{12} + \zeta^{21}) \end{aligned} \quad (1.39)$$

One can also show that **the scalar product of two spinors of rank two is proportional to the scalar product of two 3-vectors.**

$$-\frac{1}{2}\zeta^{\alpha\beta}\chi_{\alpha\beta} = \mathbf{a} \cdot \mathbf{b} \quad (1.40)$$

The relation between spinors and vectors of higher rank can be derived by noting that higher rank tensors can be constructed by multiplication of vectors.

2 Four-dimentional Vectors and Lorentz Transformation

2.1 Four-dimensional Space and Lorentz Transformation

In the theory of relativity, an **event** is described by the place where it occurs and the time when it occurred. **An event is thus represented by a point in a four-dimensional space, where one axis marks the time and the other three axes mark the space coordinate.** The coordinate of the event, (ct, x, y, z) , can also be considered as the components of the **four-dimentional radius vector**, x^μ ,

$$x^0 = ct, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z \quad (2.1)$$

We use Greek letters $\mu, \nu, \dots$ to denote the indices of four-dimentional vectors and tensors. The **interval** between two events ds is defined as

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2 \quad (2.2)$$

If we define the **metric tensor** in the 4D space as

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (2.3)$$

then

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (2.4)$$

corresponds to the “**distance**” in the 4D space.

The principle of relativity states that **the interval between two events is invariant when switched from one inertial frame of reference to another**. Since the interval can be viewed as the distance in the 4D space, the transformation of (ct, x, y, z) when switching between inertial frames of reference must leave the distance in 4D space unchanged. Such transformations could be a translation or a rotation in 4D space. The translation only leads to a shift of the origin of space coordinate or time, which is of no interest. Thus the required transformation must be a rotation in 4D space. Every rotation in the 4D can be resolved in to six rotations in xy , xz , yz , tx , ty , and tz planes. Among them, rotations in xy , xz and yz planes are just ordinary 3D spatial rotations. Therefore, we conclude that **the transformation of (ct, x, y, z) when switching between inertial frames of reference must be a rotation in 4D space which affects the t -axis**.

Consider two inertial reference systems K and K' , where K' is **moving with velocity V along the x -axis relative to K** . Clearly, the transformation between K and K' is a rotation in the tx plane. Let ϕ be the “rotation angle” of x' -axis relative to x -axis, then

$$x = x' \cosh \phi + ct' \sinh \phi, \quad ct = x' \sinh \phi + ct' \cosh \phi \quad (2.5)$$

Note that for rotations in tx , ty , and tz planes, the trigonometric functions are replaced by hyperbolic functions because of the metric in 4D space is different than in the Euclidean space.

The “rotation angle” ϕ must be related to the relative velocity V . Consider the motion of the origin of K' system in K system. Setting $x' = 0$ in the equation above, we have

$$x = ct' \sinh \phi, \quad ct = ct' \cosh \phi, \quad \frac{x}{ct} = \tanh \phi \quad (2.6)$$

Here x is the coordinate of the origin of K' system in K system and t is the time measured in K system. Clearly $x/t = V$. Thus

$$\tanh \phi = \frac{V}{c} \quad (2.7)$$

and the transformation formula becomes

$$x = \frac{x' + Vt'}{\sqrt{1 - \frac{V^2}{c^2}}}, \quad t = \frac{t' + \frac{V}{c^2}x'}{\sqrt{1 - \frac{V^2}{c^2}}} \quad (2.8)$$

It can be written in matrix form as

$$\begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-\frac{V^2}{c^2}}} & \frac{\frac{V}{c}}{\sqrt{1-\frac{V^2}{c^2}}} & 0 & 0 \\ \frac{\frac{V}{c}}{\sqrt{1-\frac{V^2}{c^2}}} & \frac{1}{\sqrt{1-\frac{V^2}{c^2}}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} \quad (2.9)$$

This is the formula of **Lorentz transformation**. The inverse transformation is

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-\frac{V^2}{c^2}}} & \frac{-\frac{V}{c}}{\sqrt{1-\frac{V^2}{c^2}}} & 0 & 0 \\ \frac{-\frac{V}{c}}{\sqrt{1-\frac{V^2}{c^2}}} & \frac{1}{\sqrt{1-\frac{V^2}{c^2}}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \quad (2.10)$$

which can be obtained by changing V to $-V$ in Eq. (2.9). The transformation matrix for rotations in ty and tz planes can be derived in the same mannar. **All rotations in the 4D space form a group named proper Lorentz group. The 3D spatial rotation group is its subgroup.**

2.2 Four-dimensional Vectors and Tensors

A quantity which has four components A^0, A^1, A^2, A^3 and is transformed as the four-dimentional radius vector (ct, x, y, z) under four-dimentional rotation is a called a **four-dimentional vector** or **4-vector**. Specifically, for the Lorentz transformation (rotation in tx plane), the 4-vector (A^0, A^1, A^2, A^3) is transformed as

$$\begin{pmatrix} A^0 \\ A^1 \\ A^2 \\ A^3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-\frac{V^2}{c^2}}} & \frac{\frac{V}{c}}{\sqrt{1-\frac{V^2}{c^2}}} & 0 & 0 \\ \frac{\frac{V}{c}}{\sqrt{1-\frac{V^2}{c^2}}} & \frac{1}{\sqrt{1-\frac{V^2}{c^2}}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A'^0 \\ A'^1 \\ A'^2 \\ A'^3 \end{pmatrix} \quad (2.11)$$

A^μ denotes the **contravariant components** of the 4-vector whereas A_μ denotes the **covariant components**. They are related to each other via

$$A_\mu = g_{\mu\nu} A^\nu, \quad A^\mu = g^{\mu\nu} A_\nu \quad (2.12)$$

Explicitly written,

$$\boxed{A_0 = A^0, \quad A_1 = -A^1, \quad A_2 = -A^2, \quad A_3 = -A^3,} \quad (2.13)$$

Here $g_{\mu\nu}$ and $g^{\mu\nu}$ are covariant and contravariant components of the metric tensor, $g_{\mu\nu} = g^{\mu\nu}$. The transformation formula of A_μ under Lorentz transformation is

$$\begin{pmatrix} A_0 \\ A_1 \\ A_2 \\ A_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} & \frac{-\frac{v}{c}}{\sqrt{1-\frac{v^2}{c^2}}} & 0 & 0 \\ \frac{-\frac{v}{c}}{\sqrt{1-\frac{v^2}{c^2}}} & \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A'_0 \\ A'_1 \\ A'_2 \\ A'_3 \end{pmatrix} \quad (2.14)$$

Note the change of sign of some elements in the transformation matrix compared to that in Eq. (2.11).

Analogous to the 4D radius vector, A^0 is called the **time component** of the 4-vector, and A^1, A^2, A^3 are called the **space components** of the 4-vector. (A^1, A^2, A^3) is transformed as an **ordinary 3D vector under purely spatial rotation**, they can thus be denoted as a 3D vector $\mathbf{A}$. We often write A^μ and A_μ as[†]

$$\boxed{A^\mu = (A^0, \mathbf{A}), \quad A_\mu = (A^0, -\mathbf{A})} \quad (2.15)$$

$A^\mu B_\mu$ is the scalar product of two 4-vectors, $A^\mu B_\mu = A_\mu B^\mu$. This product is a **four-dimensional scalar** and is invariant under 4D rotation.

A **four-dimensional tensor of rank two** is a quantity with 16 components $A^{\mu\nu}$ which are transformed as the product of the components of two 4-vectors $B^\mu C^\nu$. $A^{\mu\nu}$ denotes the contravariant components of the 4-tensor. The covariant and mix components can be obtained via

$$A_{\mu\nu} = g_{\mu\lambda} g_{\nu\sigma} A^{\lambda\sigma}, \quad A^\mu{}_\nu = g_{\nu\sigma} A^{\mu\sigma}, \quad A_\mu{}^\nu = g_{\mu\lambda} A^{\lambda\nu} \quad (2.16)$$

If $A^{\mu\nu} = A^{\nu\mu}$, then $A^{\mu\nu}$ is a **symmetrical tensor**; whereas $A^{\mu\nu} = -A^{\nu\mu}$, then $A^{\mu\nu}$ is an **anti-symmetrical tensor**. In an antisymmetrical tensor, the diagonal elements must be zero. $A^\mu{}_\mu$ is the **trace** of a 4-tensor, which is a 4-scalar. $A^\mu{}_\mu = A_\mu{}^\mu$.

The space components of an antisymmetrical 4-tensor form an antisymmetrical 3-tensor, which are transformed as an ordinary tensor under 3D spatial rotation. We denote the components of this 3-tensor as A_{ij} (see footnote). An arbitrary antisymmetrical 3-tensor can be written in terms of two vectors as

$$A_{ij} = B_i C_j - B_j C_i \quad (2.17)$$

The cross product of these two vectors, $\mathbf{A} = \mathbf{B} \times \mathbf{C}$, can be expressed using tensor notation as

$$A_i = e_{ijk} B_j C_k = -e_{ijk} C_j B_k \quad (2.18)$$

where e_{ijk} is the **three-dimensional completely antisymmetrical unit tensor of rank 3**. Its components change sign under interchange any pairs of indices. Its nonzero components equal to

[†]It's not necessary to distinguish contravariant and covariant components of a 3D vector or 3D tensor.

$(-1)^p$ where p is the number of iterations needed to rearrange ijk to 123. The components of the tensor A_{ij} and the vector $\mathbf{A}$ are related via

$$A_i = \frac{1}{2} e_{ijk} A_{jk} \quad (2.19)$$

We say the 3-vector $\mathbf{A}$ is **dual** to the antisymmetrical 3-tensor.

Now consider the **inversion of the 3D space**. Such operations can not be resolved as 3D spatial rotation and thus not an element in the 3D rotation group or in the proper Lorentz group. **The components of a vector change sign under spatial inversion.** These vectors are called **polar vectors**. The cross product of two polar vectors is an **axial vector**. **The components of an axial vector don't change sign under spatial inversion.** The axial vector is thus referred to as a **pseudovector**. The dot product of a polar vector and an axial vector is a **pseudoscalar**, which changes sign under spatial inversion. Similarly, e_{ijk} is a **pseudotensor**.

The components A^{01}, A^{02}, A^{03} of an antisymmetrical 4-tensor are transformed as a vector under spatial rotation, and changes sign under spatial inversion. They can thus be viewed as components of a polar vector. Thus, we conclude that **an antisymmetrical 4-tensor can be expressed in terms of the components of a polar vector $\mathbf{p}$ and an axial vector $\mathbf{a}$** , as

$$A^{\mu\nu} = (\mathbf{p}, \mathbf{a}) \equiv \begin{pmatrix} 0 & p_x & p_y & p_z \\ -p_x & 0 & -a_z & a_y \\ -p_y & a_z & 0 & -a_x \\ -p_z & -a_y & a_x & 0 \end{pmatrix} \quad (2.20)$$

and the covariant components are

$$A_{\mu\nu} = (-\mathbf{p}, \mathbf{a}) \quad (2.21)$$

2.3 Common 4-Vectors and 4-Tensors

In this section, we summarize some common 4-vectors and 4-tensors widely used in classical relativistic mechanics and electrodynamics. Only the contravariant components (except for the gradient) will be given, because the covariant components can be derived accordingly.

- Four-dimensional radius vector:

$$x^\mu = (ct, \mathbf{r}) \quad (2.22)$$

- Four-dimensional gradient:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, \nabla \right), \quad \partial^\mu \equiv \frac{\partial}{\partial x_\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\nabla \right) \quad (2.23)$$

- Four-dimensional velocity:

$$u^\mu \equiv \frac{dx^\mu}{ds} = \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}, \frac{\mathbf{v}}{c\sqrt{1 - \frac{v^2}{c^2}}} \right) \quad (2.24)$$

where $\mathbf{v}$ is the 3D velocity of the particle.

- Four-dimensional momentum:

$$p^\mu = \left(\frac{E}{c}, \mathbf{p} \right) \quad (2.25)$$

where $\mathbf{p}$ is the 3D momentum vector, and E is the energy or Hamiltonian function of the particle.

$$E = c\sqrt{p^2 + m^2c^2} \quad (2.26)$$

- Four-dimensional antisymmetrical angular momentum tensor:

$$M^{\mu\nu} \equiv x^\mu p^\nu - x^\nu p^\mu = \left(c\mathbf{r} \times \mathbf{p} - \frac{E\mathbf{r}}{c}, -\mathbf{M} \right) \quad (2.27)$$

$\mathbf{M} = \mathbf{r} \times \mathbf{p}$ is the 3D angular momentum, an axial vector.

- Four-dimensional potential of electromagnetic field (in SI unit):

$$A^\mu = \left(\frac{\varphi}{c}, \mathbf{A} \right) \quad (2.28)$$

φ is the scalar potential and $\mathbf{A}$ is the vector potential.

- Four-dimensional canonical momentum of a particle in external field (in SI unit):

$$P^\mu = p^\mu + qA^\mu = \left(\frac{H}{c}, \mathbf{P} \right) \quad (2.29)$$

$$\mathbf{P} = \mathbf{p} + q\mathbf{A}, \quad H = \sqrt{c^2(\mathbf{P} - q\mathbf{A})^2 + m^2c^4} + q\varphi \quad (2.30)$$

q is the charge of the particle.

- Four-dimensional antisymmetrical electromagnetic field tensor (in SI unit):

$$F^{\mu\nu} \equiv \partial^\mu A^\nu - \partial^\nu A^\mu = \left(-\frac{\mathbf{E}}{c}, \mathbf{B} \right) \quad (2.31)$$

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A} \quad (2.32)$$

- Four-dimensional current (in SI unit):

$$j^\mu = (c\rho, \mathbf{j}) \quad (2.33)$$

ρ is the charge density and $\mathbf{j}$ is the 3D current density.

In Gaussian unit, Eqs. (2.28) and (2.29) becomes

$$A^\mu = (\varphi, \mathbf{A}), \quad P^\mu = p^\mu + \frac{q}{c}A^\mu \quad (2.34)$$

while the definition of $F^{\mu\nu}$ in Eq. (2.31) remains the same. However, the definitions of the 3D electric and magnetic fields in Gaussian unit change to

$$\mathbf{E} = -\nabla\phi - \frac{1}{c}\frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A} \quad (2.35)$$

Thus in Gaussian unit,

$$F^{\mu\nu} = (-\mathbf{E}, \mathbf{B}) \quad (2.36)$$

3 Four-dimentional Spinors

3.1 Four-dimensional Spinors

In non-relativistic theory, the wave function of a particle with spin s is described by a symmetrical 3-spinor of rank $2s$. In relativistic theory, this description is still valid **in the rest frame of the particle**. However, the 3-spinor theory doesn't provides the transformation of the wave functions under Lorentz transformation. The theory of four-dimsntional spinors (4-spinors) is the extension of the 3-spinor theory.

A quantity ξ with two components, ξ^1 and ξ^2 , which are transformed as

$$\begin{pmatrix} \xi'^1 \\ \xi'^2 \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \xi^1 \\ \xi^2 \end{pmatrix} \quad (3.1)$$

with a constraint

$$\alpha\delta - \beta\gamma = 1 \quad (3.2)$$

under **transformations belonging to the proper Lorentz group** is called a **four-dimensional spinor of rank one** (4-spinor of rank one). ξ^1 and ξ^2 are the contravariant components of the 4-spinor. The covariant components, ξ_1 and ξ_2 , can be obtained via

$$\xi_\alpha = g_{\alpha\beta} \xi^\beta \quad (3.3)$$

where $g_{\alpha\beta}$ is the “metric spinor”,

$$g_{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (3.4)$$

Therefore,

$$\boxed{\xi_1 = \xi^2, \quad \xi_2 = -\xi^1} \quad (3.5)$$

The covariant components are transformed as

$$\begin{pmatrix} \xi'_1 \\ \xi'_2 \end{pmatrix} = \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \quad (3.6)$$

$\xi^\alpha \eta_\alpha$ is the scalar product of two 4-spinors, which is invariant under Lorentz transformation. $\xi^\alpha \eta_\alpha = -\xi_\alpha \eta^\alpha$. Up to now, the formulae for 4-spinors are the same for 3-spinors.

The complex conjugate of a 4-spinor ξ^* is transformed as

$$\begin{pmatrix} \xi'^{1*} \\ \xi'^{2*} \end{pmatrix} = \begin{pmatrix} \alpha^* & \beta^* \\ \gamma^* & \delta^* \end{pmatrix} \begin{pmatrix} \xi^{1*} \\ \xi^{2*} \end{pmatrix} \quad (3.7)$$

For 3-spinors, $\xi^1 \xi^{1*} + \xi^2 \xi^{2*}$ corresponds to the probability density of the particle and thus is invariant under spatial rotation. Thus $\xi^1 \xi^{1*} + \xi^2 \xi^{2*}$ must be a 3-scalar and $\xi^{\alpha*}$ must be transformed as ξ_α , which leads to an extra constraint Eq. (1.19). **However, in relativistic theory, the particle density is not a scalar but the time component of a 4-vector (4-current). It is thus**

not invariant under Lorentz transformation. Therefore, Eq. (1.19) does not hold for 4-spinors. **The transformation matrix for 4-spinors is not unitary.** It contains 6 independent parameters: the real and imaginary parts of $\alpha, \beta, \gamma, \delta$ are 8 parameters, while Eq. (3.2) gives two constraints for real and imaginary parts, respectively. This is in accordance with the 6 independent rotations in the 6 planes in 4D space.

The discussion above suggests that $\xi^{\alpha*}$ is not transformed as ξ^α nor ξ_α . **There are thus two types of spinors in relativistic theory that transform differently.** We use dotted indices $\dot{\alpha}, \dot{\beta}, \dots$ to label the components of the second type of spinor, which is transformed as $\xi^{\alpha*}$:

$$\boxed{\eta^{\dot{\alpha}} \sim \xi^{\alpha*}} \quad (3.8)$$

The covariant components are given by $\eta_{\dot{\alpha}} = g_{\dot{\alpha}\dot{\beta}}\eta^{\dot{\beta}}$ where $g_{\dot{\alpha}\dot{\beta}} = g_{\alpha\beta}$. Thus

$$\boxed{\eta_{\dot{1}} = \eta^{\dot{2}}, \quad \eta_{\dot{2}} = -\eta^{\dot{1}}} \quad (3.9)$$

The behavior of 4-spinors under spatial rotation should be the same as 3-spinors where the complex conjugate of covariant components are transformed as contravariant components and *vice versa*, thus

$$\boxed{\eta_{\dot{\alpha}} \sim \xi^\alpha \quad \text{under 3D spatial rotations}} \quad (3.10)$$

The wave function of a spin $\frac{1}{2}$ particle in its rest frame is described by a 3-spinor. This 3-spinor could have two “four-dimensional origins”, i.e., $\eta_{\dot{\alpha}}$ or ξ^α . Both $\eta_{\dot{\alpha}}$ and ξ^α are needed to describe of the particle’s wave function in an arbitrary frame of reference.

4-spinors of higher rank are defined as sets of quantities which are transformed as products of the components of spinors of rank one. Their indices could be partially dotted and partially undotted. For example, there are **three types of 4-spinors of rank two**:

$$\zeta^{\alpha\beta} \sim \xi^\alpha \Xi^\beta, \quad \zeta^{\alpha\dot{\beta}} \sim \xi^\alpha \eta^{\dot{\beta}}, \quad \zeta^{\dot{\alpha}\dot{\beta}} \sim \eta^{\dot{\alpha}} H^{\dot{\beta}} \quad (3.11)$$

Therefore, a pair of numbers (k, l) are needed to specify the numbers of dotted and undotted indices in a 4-spinor of rank $k + l$. The contraction of a 4-spinor or product of 4-spinors can only be done with respect to the indices of the same kind. The definitions of symmetrical and antisymmetrical 4-spinors follow the similar rule. **A symmetrical 4-spinor of rank (k, l) has $(k + 1)(l + 1)$ independent components.** Note that unlike 3-spinors, not all symmetrical 4-spinors of rank $k + l = 2s$ have $2s + 1$ independent components.

3.2 Relation between 4-Spinor and 4-Vectors

The 4-spinor of rank $(1, 1)$ $\zeta^{\alpha\dot{\beta}}$ has $2 \times 2 = 4$ independent components, same as a 4-vector. Analogous to the three-dimensional case, **there must be a correspondence between the components of $\zeta^{\alpha\dot{\beta}}$ and the 4-vector $a^\mu = (a^0, \mathbf{a})$.**

First we note that under spatial rotation, $\zeta^{\alpha\dot{\beta}}$ is transformed as the 3-spinor ζ^α_β , whose components are related to the components of the 3-vector $\mathbf{a}$ via Eq. (1.39). Thus

$$\begin{aligned} a^1 &= \frac{1}{2}(\zeta^{1\dot{2}} + \zeta^{2\dot{1}}) \\ a^2 &= \frac{i}{2}(\zeta^{1\dot{2}} - \zeta^{2\dot{1}}) \\ a^3 &= \frac{1}{2}(\zeta^{1\dot{1}} - \zeta^{2\dot{2}}) \end{aligned} \quad (3.12)$$

For the relation with a^0 , we note that the expression $\xi^1\xi^{1*} + \xi^2\xi^{2*}$ corresponds to the probability density of the particle, which is the time component of the 4-current. Thus

$$a^0 \sim \zeta^{1\dot{1}} + \zeta^{2\dot{2}} \quad (3.13)$$

To determine the coefficient, we can generalize the relation in Eq. (1.40) to 4D space, so that

$$\boxed{\zeta^{\alpha\dot{\beta}}\zeta_{\alpha\dot{\beta}} = 2a^\mu a_\mu} \quad (3.14)$$

We then obtain the correspondence formulae

$$\boxed{\begin{aligned} a^0 &= \frac{1}{2}(\zeta^{1\dot{1}} + \zeta^{2\dot{2}}) \\ a^1 &= \frac{1}{2}(\zeta^{1\dot{2}} + \zeta^{2\dot{1}}) \\ a^2 &= \frac{i}{2}(\zeta^{1\dot{2}} - \zeta^{2\dot{1}}) \\ a^3 &= \frac{1}{2}(\zeta^{1\dot{1}} - \zeta^{2\dot{2}}) \end{aligned}} \quad (3.15)$$

The inverse formulae are

$$\boxed{\zeta^{\alpha\dot{\beta}} = \begin{pmatrix} a^0 + a^3 & a^1 - ia^2 \\ a^1 + ia^2 & a^0 - a^3 \end{pmatrix} = a^0 \mathbf{I} + \boldsymbol{\sigma} \cdot \mathbf{a}} \quad (3.16)$$

where $\mathbf{I}$ is the 2×2 identity matrix and $\boldsymbol{\sigma}$ are the Pauli matrices. Eq. (3.15) can also be written in terms of Pauli matrices as

$$\boxed{a^0 = \frac{1}{2}\text{Tr } \zeta, \quad \mathbf{a} = \frac{1}{2}\text{Tr}(\zeta \boldsymbol{\sigma})} \quad (3.17)$$

The multiplication of ζ and Pauli matrices follows the ordinary matrix multiplication rule.

In general, a symmetrical 4-spinor of rank (k, k) is equivalent to an irreducible symmetrical 4-tensor of rank k .

3.3 Representation of the Proper Lorentz Group

For 3D spatial rotation, both the 3×3 rotation matrices and the 2×2 unitary and unimodular matrices for the binary transformation of spinors are (irreducible) representations of the 3D rotation group. Similarly, an element in the proper Lorentz group can be represented by a 4×4 matrix

given in Eq. (2.9), or a 2×2 unimodular matrix for binary transformation of 4-spinors. In this section, we will derive the explicit expression of these 2×2 unimodular matrices using the relation between 4-spinors and 4-vectors.

Let $\mathbf{B}$ be the transformation matrix for 4-spinors

$$\mathbf{B} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \xi'^\alpha = (\mathbf{B}\xi)^\alpha \quad (3.18)$$

For a 4-spinor of rank (1,1), $\zeta^{\alpha\dot{\beta}}$, its transformation can be written in matrix form as

$$\zeta' = \mathbf{B}\zeta\mathbf{B}^H \quad (3.19)$$

The multiplication of the 4-spinor of rank (1,1) ζ with matrix $\mathbf{B}$ follows the ordinary matrix multiplication rule. $\mathbf{B}^H$ is the Hermitian adjoint of $\mathbf{B}$. For an infinitesimal transformation, we approximate $\mathbf{B}$ to the first order, $\mathbf{B} \approx \mathbf{I} + \boldsymbol{\lambda}$, then the first-order approximation of ζ' is

$$\zeta' = \zeta + (\boldsymbol{\lambda}\zeta + \zeta\boldsymbol{\lambda}^H) \quad (3.20)$$

Now consider a Lorentz transformation to a frame of reference moving with an infinitesimal relative velocity $\delta\mathbf{V}$, from Eq. (2.11), the 4-vector $A^\mu = (a^0, \mathbf{a})$ is transformed to

$$a'^0 = a^0 - \frac{\delta\mathbf{V}}{c} \cdot \mathbf{a}, \quad \mathbf{a}' = \mathbf{a} - a^0 \frac{\delta\mathbf{V}}{c} \quad (3.21)$$

Using Eq. (3.17),

$$a'^0 = a^0 - \frac{1}{2}\text{Tr} \left[\zeta \left(\boldsymbol{\sigma} \cdot \frac{\delta\mathbf{V}}{c} \right) \right], \quad \mathbf{a}' = \mathbf{a} - \frac{1}{2}\text{Tr} \left(\zeta \frac{\delta\mathbf{V}}{c} \right) \quad (3.22)$$

On the other hand,

$$a'^0 = \frac{1}{2}\text{Tr} \zeta' = a^0 + \frac{1}{2}\text{Tr} (\boldsymbol{\lambda}\zeta + \zeta\boldsymbol{\lambda}^H) = a^0 + \frac{1}{2}\text{Tr} [\zeta (\boldsymbol{\lambda} + \boldsymbol{\lambda}^H)] \quad (3.23a)$$

$$\mathbf{a}' = \frac{1}{2}\text{Tr}(\zeta'\boldsymbol{\sigma}) = \mathbf{a} + \frac{1}{2}\text{Tr} [(\boldsymbol{\lambda}\zeta + \zeta\boldsymbol{\lambda}^H)\boldsymbol{\sigma}] = \mathbf{a} + \frac{1}{2}\text{Tr} [\zeta (\boldsymbol{\sigma}\boldsymbol{\lambda} + \boldsymbol{\lambda}^H\boldsymbol{\sigma})] \quad (3.23b)$$

Thus

$$\boldsymbol{\lambda} + \boldsymbol{\lambda}^H = -\boldsymbol{\sigma} \cdot \frac{\delta\mathbf{V}}{c} \quad (3.24a)$$

$$\boldsymbol{\sigma}\boldsymbol{\lambda} + \boldsymbol{\lambda}^H\boldsymbol{\sigma} = -\frac{\delta\mathbf{V}}{c} \quad (3.24b)$$

Solving for $\boldsymbol{\lambda}$, we have

$$\boldsymbol{\lambda} = \boldsymbol{\lambda}^H = -\frac{1}{2}\boldsymbol{\sigma} \cdot \mathbf{n}_V \frac{\delta V}{c} \quad (3.25)$$

where $\mathbf{n}_V$ is the unit vector in the direction of $\delta\mathbf{V}$. Recall that this Lorentz transformation corresponds to a rotation in the plane spanned by t and $\mathbf{n}_V$ by $\delta\phi$ where

$$\tanh \delta\phi \approx \delta\phi = \frac{\delta V}{c} \quad (3.26)$$

Thus

$$\mathbf{B}_{\text{Lorentz}} = \mathbf{I} - \frac{1}{2} \boldsymbol{\sigma} \cdot \mathbf{n}_V \delta\phi \quad (3.27)$$

The transformation to a frame moving with a finite velocity corresponds to a rotation by a finite angle ϕ , repeating the infinitesimal transformation by $\phi/\delta\phi$ times, we have

$$\boxed{\mathbf{B}_{\text{Lorentz}} = \exp\left(-\frac{1}{2} \phi \boldsymbol{\sigma} \cdot \mathbf{n}_V\right) = \cosh \frac{\phi}{2} - \boldsymbol{\sigma} \cdot \mathbf{n}_V \sinh \frac{\phi}{2}} \quad (3.28)$$

and $\tanh \phi = \frac{V}{c}$. $\mathbf{V}$ is the velocity of K' system relative to K . $\mathbf{B}_{\text{Lorentz}}$ is **Hermitian**.

The matrix for ordinary 3D spatial rotation can be derived in the same way. The result is already given in Eq. (1.34), which can be written using Pauli matrices as

$$\boxed{\mathbf{B}_{\text{rot}} = \exp\left(\frac{1}{2} i \phi \boldsymbol{\sigma} \cdot \mathbf{n}\right) = \cos \frac{\phi}{2} + i \boldsymbol{\sigma} \cdot \mathbf{n} \sin \frac{\phi}{2}} \quad (3.29)$$

Here the coordinate system is rotated counterclockwise by ϕ around the unit vector $\mathbf{n}$. $\mathbf{B}_{\text{rot}}$ is **unitary**.

4 Inversion of Spinors

4.1 Inversion of 3-Spinors

In relativistic quantum theory, the transformation of quantities under **spatial inversion (parity transformation)** is of great importance. An inversion operation can not be resolved as rotations and is thus not an element in the 3D rotation group or the proper Lorentz group. Spatial inversion together with all 4D rotations form the **extended Lorentz group**.

First consider the transformation of 3-spinors in non-relativistic theory. In 3D space, the spin vector, as a type of angular momentum, is an axial vector. Its components s_x, s_y, s_z are thus unchanged under spatial inversion. Recall that the wave function of a spin $\frac{1}{2}$ particle has two components labeled by $\sigma \equiv s_z/\hbar$. Since s_z is unchanged, **the component ψ_σ must be transformed into a multiple of itself**:

$$\psi_\sigma \rightarrow P \psi_\sigma \quad (4.1)$$

where P is a constant factor called the **parity**. By two consecutive inversions we revert back to the original coordinate system. This can be viewed as a rotation by $\phi = 0$ or 2π . These cases are not equivalent for spinors because a rotation by 2π changes the sign of a spinor. For the $\phi = 0$ case, we have

$$P^2 = 1, \quad P = \pm 1 \quad (4.2)$$

For the $\phi = 2\pi$ case, we have

$$P^2 = -1, \quad P = \pm i \quad (4.3)$$

In practice, we can use either Eqs. (4.2) or (4.3) to define the parity of a 3-spinor. But the choice must be consistent for all spinors. Furthermore, the assignment of a particular parity (for example,

$+i$ or $-i$) to a spinor has no absolute significance, because an inversion can always be performed together with a rotation by 2π , which changes the sign of a spinor. However, the **relative parity** of two spinors, defined as the parity of the scalar $\xi^\alpha \eta_\alpha$ has absolute significance, because rotation by 2π changes the sign of both spinors thus not affect the parity of the scalar.

A 3-spinor can have a definite parity because in 3D space, inversion commutes with all 3D spatial rotations. Its matrix representation thus must be proportional to the identity matrix. The scenario is quite different in 4D space, as **the spatial inversion does not commute with Lorentz transformations:**

$$\hat{P}\hat{L}(\mathbf{V}) = \hat{L}(-\mathbf{V})\hat{P} \quad (4.4)$$

where $\hat{P}$ denotes the inversion transformation, $\hat{L}(\mathbf{V})$ denotes the Lorentz transformation to a frame of reference moving with a relative velocity $\mathbf{V}$.

4.2 Inversion of 4-Spinors

Based on the discussion in the previous section, **the inversion can not transform a 4-spinor ξ^α into a multiple of itself, but into an expression involving other quantities.** Since in 4D space, the inversion still commutes with spatial rotations, the transformed quantity must be transformed as ξ^α under space rotation. Such quantities could be ξ'^α or $\eta_{\dot{\alpha}}$. The transformation from ξ^α to ξ'^α would be a spatial rotation. Thus, **ξ^α can only be transformed into $\eta_{\dot{\alpha}}$ under inversion and vice versa:**

$$\xi^\alpha \rightarrow P\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow P\xi^\alpha \quad (4.5)$$

Similar to the 3D case, by two consecutive inversions we revert back to the original coordinate system, which can be viewed as a rotation by $\phi = 0$ or 2π . Thus $P^2 = \pm 1$.

If we take $P^2 = 1$, then

$$\xi^\alpha \rightarrow \eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow \xi^\alpha \quad (4.6a)$$

$$\xi_\alpha \rightarrow -\eta^{\dot{\alpha}}, \quad \eta^{\dot{\alpha}} \rightarrow -\xi_\alpha \quad (4.6b)$$

The transformation of covariant components comes from Eqs. (3.5) and (3.9). Let $H^{\dot{\alpha}} = \xi^{*\alpha}$ and $\Xi_\alpha = \eta_{\dot{\alpha}}^*$, then they are transformed as

$$\Xi_\alpha \rightarrow H^{\dot{\alpha}}, \quad H^{\dot{\alpha}} \rightarrow \Xi_\alpha \quad (4.7)$$

which differs in sign from the transformation of ξ_α and $\eta^{\dot{\alpha}}$.

Alternatively, if we take $P^2 = -1$, then

$$\boxed{\xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha} \quad (4.8a)$$

$$\boxed{\xi_\alpha \rightarrow -i\eta^{\dot{\alpha}}, \quad \eta^{\dot{\alpha}} \rightarrow -i\xi_\alpha} \quad (4.8b)$$

The transformation rule for the complex conjugates $H^{\dot{\alpha}} = \xi^{*\alpha}$ and $\Xi_\alpha = \eta_{\dot{\alpha}}^*$ are now the same as the transformation of ξ_α and $\eta^{\dot{\alpha}}$:

$$\Xi_\alpha \rightarrow -iH^{\dot{\alpha}}, \quad H^{\dot{\alpha}} \rightarrow -i\Xi_\alpha \quad (4.9)$$

We will follow the rules defined in Eq. (4.8) from hereon.

The quantity $\xi^\alpha \pm \eta_{\dot{\alpha}}$ has a definite parity $\pm i$. It is transformed as a spinor under spatial rotation but not under Lorentz transformations. It is thus not a 4-spinor.

In the extended Lorentz group, a single 4-spinor is insufficient to describe the state of a spin $\frac{1}{2}$ particle under all symmetry transformations. We must simultaneously consider a pair of spinors, $(\xi^\alpha, \eta_{\dot{\alpha}})$, which is called a **bispinor of rank one**. If we denote the bispinor as

$$\psi \equiv \begin{pmatrix} \xi \\ \eta \end{pmatrix} \equiv \begin{pmatrix} \xi^1 \\ \xi^2 \\ \eta_1 \\ \eta_2 \end{pmatrix} \quad (4.10)$$

then we may write the matrix representation of inversion as

$$\mathbf{P} = \begin{pmatrix} \mathbf{0} & i\mathbf{I} \\ i\mathbf{I} & \mathbf{0} \end{pmatrix} \quad (4.11)$$

thus

$$\mathbf{P} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} i\eta \\ i\xi \end{pmatrix} \quad (4.12)$$

In this representation, the matrices for spatial rotation and Lorentz transformation have the form

$$\tilde{\mathbf{B}}_{\text{rot}} = \begin{pmatrix} \mathbf{B}_{\text{rot}} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{\text{rot}} \end{pmatrix} \quad (4.13)$$

$$\tilde{\mathbf{B}}_{\text{Lorentz}} = \begin{pmatrix} \mathbf{B}_{\text{Lorentz}} & \mathbf{0} \\ \mathbf{0} & \mathbf{g}\mathbf{B}_{\text{Lorentz}}^*\mathbf{g}^T \end{pmatrix} \quad (4.14)$$

where $\mathbf{B}_{\text{rot}}$ and $\mathbf{B}_{\text{Lorentz}}$ are the binary transformation matrices given in Eqs. (3.28) and (3.29). $\mathbf{g}$ is the matrix form of the metric spinor and $\mathbf{g}^T$ is its transpose. Note that $\mathbf{g}\mathbf{B}_{\text{rot}}^*\mathbf{g}^T = \mathbf{B}_{\text{rot}}$. Clearly $\mathbf{P}$ commutes with $\tilde{\mathbf{B}}_{\text{rot}}$ but not with $\tilde{\mathbf{B}}_{\text{Lorentz}}$. **These 4×4 matrices are irreducible representations of the extended Lorentz group, but are reducible representations when only considering the proper Lorentz group.**

There are two types of scalar product of two bispinors $(\xi^\alpha, \eta_{\dot{\alpha}})$ and $(\Xi^\alpha, H_{\dot{\alpha}})$:

$$\xi^\alpha \Xi_\alpha + \eta_{\dot{\alpha}} H^{\dot{\alpha}} \quad (4.15)$$

is a true scalar, which is invariant under inversion, where as

$$\xi^\alpha \Xi_\alpha - \eta_{\dot{\alpha}} H^{\dot{\alpha}} \quad (4.16)$$

is a pseudoscalar.

4.3 Inversion of 4-Spinors of Higher Rank

Similar to the scalar product, **there are also two types of 4-spinors of rank (1,1) classified based on their transformation rules under inversion.** For the first type,

$$\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha H^{\dot{\beta}} + \Xi^\alpha \eta^{\dot{\beta}} \quad (4.17)$$

Under inversion

$$\zeta^{\alpha\dot{\beta}} \rightarrow \zeta_{\dot{\alpha}\beta} \quad (4.18)$$

This 4-spinor $\zeta^{\alpha\dot{\beta}}$ is equivalent to a 4-vector that is transformed as $(a^0, \mathbf{a}) \rightarrow (a^0, -\mathbf{a})$ under inversion. The space components of the 4-vector that this 4-spinor is equivalent to change sign, and thus form a polar vector. The 4-vector is a true vector.

For the second type,

$$\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha H^{\dot{\beta}} - \Xi^\alpha \eta^{\dot{\beta}} \quad (4.19)$$

Under inversion

$$\zeta^{\alpha\dot{\beta}} \rightarrow -\zeta_{\dot{\alpha}\beta} \quad (4.20)$$

This 4-spinor $\zeta^{\alpha\dot{\beta}}$ is equivalent to a 4-vector that is transformed as $(a^0, \mathbf{a}) \rightarrow (-a^0, \mathbf{a})$ under inversion. The space components of the 4-vector that this 4-spinor is equivalent to do not change sign, and thus form an axial vector. The 4-vector is then a pseudovector.

Symmetrical spinors of rank (2,0) or (0,2) each only has one type,

$$\zeta^{\alpha\beta} \sim \xi^\alpha \Xi^\beta + \Xi^\alpha \xi^\beta, \quad \chi_{\dot{\alpha}\dot{\beta}} \sim \eta_{\dot{\alpha}} H_{\dot{\beta}} + H_{\dot{\alpha}} \eta_{\dot{\beta}} \quad (4.21)$$

Under inversion

$$\boxed{\zeta^{\alpha\beta} \rightarrow -\chi_{\dot{\alpha}\dot{\beta}}, \quad \chi_{\dot{\alpha}\dot{\beta}} \rightarrow -\zeta^{\alpha\beta}} \quad (4.22)$$

The pair $(\zeta^{\alpha\beta}, \chi_{\dot{\alpha}\dot{\beta}})$ forms a **bispinor of rank 2**. This bispinor has $3 + 3 = 6$ independent components, same as an antisymmetrical 4-tensor $a^{\mu\nu} = (\mathbf{p}, \mathbf{a})$. Thus **there must be some correspondence between a bispinor of rank 2 and an antisymmetrical 4-tensor.**

The antisymmetrical 4-tensor $a^{\mu\nu}$ is transformed under inversion as

$$(\mathbf{p}, \mathbf{a}) \rightarrow (-\mathbf{p}, \mathbf{a}) \quad (4.23)$$

We can construct the following two quantities,

$$\mathbf{a}^2 - \mathbf{p}^2 = \frac{1}{2} a^{\mu\nu} a_{\mu\nu}, \quad \mathbf{a} \cdot \mathbf{p} = \frac{1}{8} e_{\mu\nu\lambda\sigma} a^{\mu\nu} a^{\lambda\sigma} \quad (4.24)$$

where $e_{\mu\nu\lambda\sigma}$ is the four-dimensional completely antisymmetrical unit tensor. The first quantity is a scalar and the second quantity is a pseudoscalar. Both are invariant under the proper Lorentz group. We define two complex vectors

$$\boxed{\mathbf{f}_\pm = \mathbf{p} \pm i\mathbf{a}} \quad (4.25)$$

Clearly, the norm of $\mathbf{f}_\pm$ is invariant under the proper Lorentz group. **The proper Lorentz group can thus be represented as 3D “rotation” of the complex vectors $\mathbf{f}_\pm$ with complex angles.** Under inversion, $\mathbf{f}_\pm$ is transformed as

$$\mathbf{f}_+ \rightarrow -\mathbf{f}_-, \quad \mathbf{f}_- \rightarrow -\mathbf{f}_+ \quad (4.26)$$

which is the same as the transformation rule of $\zeta^{\alpha\beta}$ and $\chi_{\dot{\alpha}\dot{\beta}}$. Note that under pure spatial rotation, $\mathbf{f}_\pm$ is transformed as an ordinary 3-vector, whereas $\zeta^{\alpha\beta}$ and $\chi_{\dot{\alpha}\dot{\beta}}$ are transformed as symmetrical 3-spinors of rank two. According to Eq. (1.39), evidently,

$$\boxed{\begin{aligned} f_x^+ &= \frac{1}{2}(\zeta^{22} - \zeta^{11}), & f_y^+ &= -\frac{i}{2}(\zeta^{11} + \zeta^{22}), & f_z^+ &= \frac{1}{2}(\zeta^{12} + \zeta^{21}) \\ f_x^- &= \frac{1}{2}(\chi_{\dot{2}\dot{2}} - \chi_{\dot{1}\dot{1}}), & f_y^- &= -\frac{i}{2}(\chi_{\dot{1}\dot{1}} + \chi_{\dot{2}\dot{2}}), & f_z^- &= \frac{1}{2}(\chi_{\dot{1}\dot{2}} + \chi_{\dot{2}\dot{1}}) \end{aligned}} \quad (4.27)$$

We summarize the correspondence among the wave function of a spin s particle, spinors and tensors as follows:

- The wave function of a spin s particle is described in its rest frame by a symmetrical 3-spinor of rank $2s$.
- A symmetrical 3-spinor of rank 2 is equivalent to a 3-vector via Eq. (1.39).
- A symmetrical 3-spinor of rank $2s$ is equivalent to an irreducible symmetrical 3-tensor of rank s .
- A 4-spinor of rank (1,1) is equivalent to a 4-vector or a 4-pseudovector via Eqs. (3.15)–(3.17).
- A symmetrical 4-spinor of rank (k, k) is equivalent to an irreducible symmetrical 4-tensor or 4-pseudotensor of rank k .
- A 4-bispinor of rank 2 is equivalent to an antisymmetrical 4-tensor of rank 2 via Eq. (4.27).